ENGR 852 Motion Estimator

V4 design, verification, and 14 nm implementation record | November 4, 2026

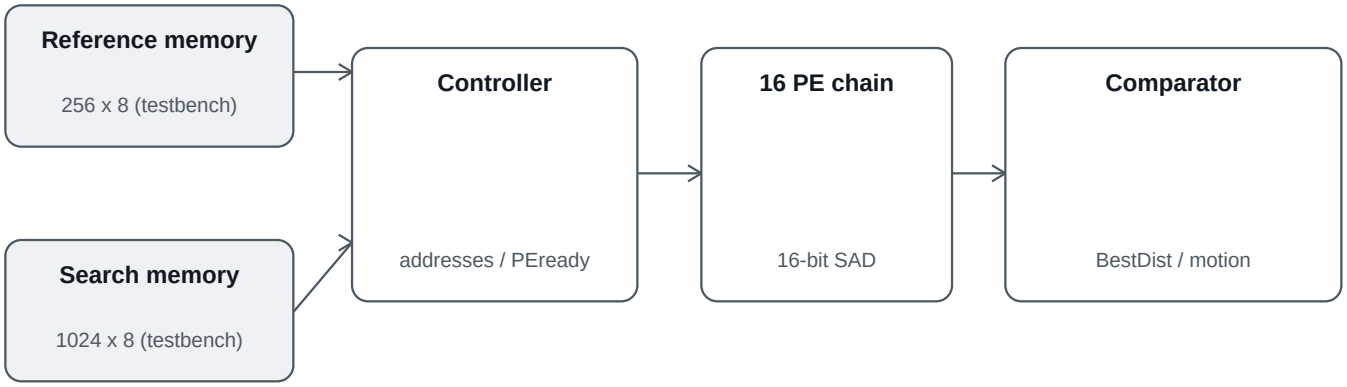
Document scope: actual VCS/Verdi verification, full-precision SAD correction, mapped SAED14 synthesis measurements, physical-flow checkpoint, and outstanding sign-off evidence. Measurements are labeled by their flow stage.

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1. Design target and memory boundary

The design compares an 8-bit grayscale **16 x 16** reference block against candidate positions in a **31 x 31** search window. A 16-PE systolic datapath performs absolute differences and accumulation while the controller generates search/reference addresses and candidate vectors. The comparator selects the minimum sum of absolute differences (SAD).

External testbench memories are NOT synthesized inside top.



Reference/search data and addresses cross the memory interface

Course setting	Required value
Clock frequency and cycle	520 MHz; 1.923 ns period
Input and output delay	0.500 ns each
Clock uncertainty	0.200 ns
Reference/search memory	Outside synthesizable top; testbench models drive R, S1, S2
SAD precision	16 bits to accommodate 256 x 255 = 65,280 (FF00)

The provided top-level RTL has memory-data inputs R/S1/S2 and address outputs AddressR/AddressS1/AddressS2; only the controller, processing elements and comparator belong in the physical chip.

2. Functional RTL validation

The original source ZIP was preserved separately. Baseline directed and randomized tests were executed with Synopsys VCS W-2024.09-1; Verdi W-2024.09-1 displayed genuine FSDB waveform data. The clock toggles every 0.9615 ns, producing the target period of 1.923 ns at timescale 1ns/100fs.

Stimulus / scenario	Observed result	Interpretation
Directed reference/search fixture	BestDist=00; X=7; Y=8; candidate count 271	Perfect match, signed vector (+7,-8)
Single-pixel +3 perturbation	BestDist=03; X=E; Y=D; candidate count 1542	Nonzero SAD, signed vector (-2,-3)
Upper-corner boundary	BestDist=00; X=7; Y=7; count 4111	Last candidate, signed vector (+7,+7)
PE / PEend / comparator directed	12 checks, 0 failures	Saturation, mux path, strict-less-than behavior
Randomized saturating-PE scoreboard	3 seeds, 24,576 PE comparisons, 0 mismatches	The original 8-bit *saturating* behavior matches its reference

Interpretation: the randomized PE scoreboard establishes agreement with the defined saturating 8-bit behavior; it does **not** prove that 8-bit SAD gives the correct minimum for every image. This motivated the separate full-precision correction.

2.1 Reproduced saturation defect

For the directed all-high-contrast fixture, all 256 reference pixels are 00 and every search pixel is FF. Each candidate has a true SAD of 256 x 255 = 65,280 (hex FF00). Original PEs saturate at 8-bit FF. The original comparator initializes BestDist to FF and updates only when newDist is **strictly smaller**, so no saturated candidate can be accepted.

Output	Original RTL / VCS observation	Corrected variant / VCS observation
BestDist	FF	FF00
motionX / motionY	X / X (unknown)	8 / 8 = signed (-8,-8)
completed	1	1
Comparator newBest	0; no valid winner	1 on first equal-SAD candidate

The 16-bit fix widens the PE accumulators, comparator datapath and best-distance register. The independent V3B hold variant also avoids output changes after completion; its directed run retained the corrected result for 13 further clock cycles. The professor source and teammate implementation were not overwritten.

3. Native waveform evidence

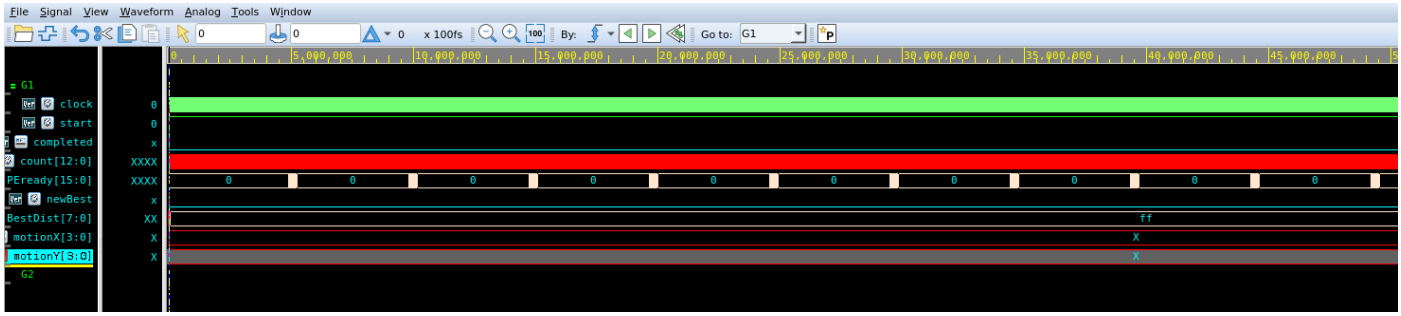


Figure 1. Original 8-bit VCS trace displayed in Verdi nWave. PEready pulses occur but newBest stays low; BestDist remains FF and motionX/Y are unknown.

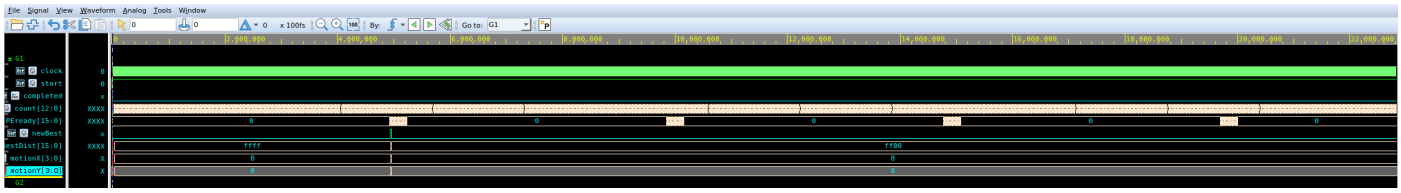


Figure 2. Corrected 16-bit Verdi nWave trace. NewBest updates BestDist to FF00, and the first candidate is encoded as 8,8 (signed -8,-8).

Both images were captured from the nWave application window without including unrelated desktop applications. Cropping was limited to the waveform panels; the original raw evidence is preserved in earlier V3 records.

As in the team controller slide, the signal list includes clock, start, completed, count[12:0], PEready[15:0], newBest, BestDist and motionX/Y. This contribution focuses on the comparator and SAD-precision failure rather than repeating the controller-row demonstration.

4. SAED14 Fusion Compiler synthesis results

The instructor-prescribed local Fusion Compiler setup loaded X-2025.06-SP2a and SAED14 RVT libraries. The example environment and motion-estimator mapped compile completed. The numbers below were read from native Synopsys reports and are **mapped-cell measurements**, not final routed core area.

Report metric	Original 8-bit RTL	Corrected 16-bit V3B
Mapped standard-cell area	881.96 um2	1,193.03 um2
Mapped cell count	1,693	1,905
Combinational cells	1,413	1,489
Sequential cells	280	416
Memory macros / black boxes	0	0
Compile-stage setup slack	First SDC run invalid input-delay collection	+0.06 ns (MET)

Corrected implementation area grows by **311.07 um2 (35.3%)** relative to the original. The 16-bit datapath preserves SAD values up to 65,280 without loss. Timing impact requires further optimization and post-route analysis.

4.1 Timing methodology

• Clock 1.923 ns and uncertainty 0.200 ns are set in the design constraints.

• External input delay and output delay are each 0.500 ns.

• An unsupported expression in the first motion-estimator input-delay command was corrected, and the reloaded timing path confirmed 0.500 ns input delay and 0.200 ns uncertainty.

• The measured +0.06 ns path is a compile-stage max-delay setup report. It is **not** a post-layout PrimeTime setup or hold measurement.

4.2 Power assumptions

The mapped flow can produce preliminary power reports, but a verified numeric final power figure and activity assumptions were not recovered in the evidence available for this report. A legitimate final power value must identify the cell library corner, clock rate, toggle/activity basis, and post-route extracted interconnect model. A nominal estimate is not inserted as a measured result.

5. Physical implementation and sign-off status

Required stage	Checkpoint	Evidence needed for final submission
Logic synthesis and mapping	PASS	Technology-mapped netlist, reports, tool scripts
Floorplan	PASS	Generated floorplan DEF, utilization/core dimensions
Placement (V3B)	PASS in reviewed log	Placed-cell layout and post-place checks
Clock-tree synthesis (V3B)	NOT VERIFIED	CTS completion log, skew and congestion/timing
Detailed routing	NOT VERIFIED	Routed layout, routing reports, final optimized netlist
Design-rule checking	NOT VERIFIED	Complete DRC report and violation count
Parasitic extraction	NOT VERIFIED	Extracted RC data, SPEF or approved equivalent
PrimeTime setup/hold	NOT VERIFIED	Post-layout max and min path reports including slack
Final layout area/power	NOT VERIFIED	Layout bounding box/core area; final power report

5.1 Clock-tree congestion diagnosed

An earlier clock-tree stage on the original 8-bit design failed with Fusion Compiler PLACE-004 / PLACE-006: **111.73% utilization**, requiring approximately **1,456 um2** in only **1,303 um2** available placement area. Placement success at a prior stage cannot establish a later CTS pass. A larger floorplan / lower target utilization and CTS rerun are indicated, followed by fresh timing, route and physical verification.

The corrected V3B placement run returned exit code 0 with no reported placement errors. Its CTS command was launched, but the final CTS report was not recovered after the SSH terminal became unresponsive. No route or sign-off results are claimed.

5.2 Minimum post-route sign-off package

• Complete placed-and-routed design (DEF/GDS or approved layout formats), gate-level Verilog and SDC.

• DRC report with explicit pass/fail and rule check summary; verified post-route area and utilization.

• Extracted parasitic data / SPEF and extraction consistency report.

• PrimeTime setup report using max delay, and hold report using min delay, each with worst path and slack.

• Power report at documented activity, voltage and process corner; full scripts and source listings.

6. Course rubric audit and reproducibility

The course 90–100 bracket requires every major design and implementation element to be correctly verified and implemented, with clear documentation. RTL and mapped results are substantive, but missing CTS, routing, DRC, extracted parasitics and post-route setup/hold mean the current result cannot responsibly be described as fully satisfying that bracket.

Course item	Current evidence	Required before claiming complete
RTL and Verilog testbench	Professor ZIP; independent directed/full-precision testbenches	Broader full-precision top-level regression if not yet done
Simulation results	VCS logs, Verdi FSDB screenshots	Retain exact seeds, failing inputs and outcomes
Synthesis / gate-level netlist	Fusion compiler mapping reports and intermediate netlist	Export optimized final post-route gate netlist
Physical layout / DRC	Floorplan and placement checkpoint	Routed GDS/DEF and complete DRC output
Parasitics and final timing	Configured technology models	Extract actual RC and PrimeTime setup + hold
Area and power	Cell area from compile stage	Final routed layout area and power with units/corner
Report and presentation	Three independent V4 slides plus 19-slide V4 deck	Insert genuine sign-off results once available

6.1 Safe continuation once SSH is authenticated

Use a new authenticated Linux terminal, not a local macOS shell. Inspect the existing V3B CTS log before rerunning or changing a stage:

```
cd ~/ENGR852_Project/16_v3b_final_asic_fc
tcsh
source scripts/synopsys_setup.tcl
tail -n 40 v3b_cts.out
grep -E "^Error:|PLACE-004|PLACE-006" v3b_cts.out | tail -20
```

After CTS is truly complete, follow the instructor flow: route, signoff and PrimeTime. Never proceed solely because a Makefile command returned code zero; check logs, layout outputs, violations, extraction and the actual max/min timing slack.

6.2 Evidence and complete source archive

The versioned repository contains the professor original source, independent Verilog/SystemVerilog checks, SDC, slide build scripts, native waveform screenshots and a timestamped engineering notebook. Full source text is stored as editable files in the repository rather than raster images or copied code screens.

• **RTL:** rtl/top.v; rtl/experimental/top_full_precision_v3.v; rtl/experimental/top_full_precision_finish_hold.v

• **Stimulus:** tb/top_tb.v; tb/top_tb_direct.v; verification/huy_v3_full_precision_tb.sv; verification/huy_v3_hold_tb.sv; verification/huy_v3_all_saturated_tb.sv

• **Constraints:** constraints/top_520mhz.sdc and the copied course Fusion scripts on the Linux workspace.

• **Reports:** docs/UPDATE-016_V3_PPA_AND_SLIDES.md and docs/UPDATE-017_V4_FINAL_CHECKPOINT.md.

• **Presentations:** deliverables/ENGR852_Huy_V4_Detailed_Waveform_3_Slides.pptx and deliverables/ENGR852_V4_Full_Team_19_Slides.pptx.

Repository: github.com/huynguyen1017/motion-estimator-asic-lab-notebook-2026

Final assessment caution: this report documents completed work and remaining implementation measurements; it is not a fabricated post-layout sign-off report.